GENERATING RADICAL MAGIC DIAMONDS

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Abstract. Here we introduce the theory of magic squares and, in particular, magic squares of squares. Using elementary number theory, we exploit a series of magic square invariants to develop a single Diophantine equation whose solutions are in correspondence with the collection of radical magic diamonds, i.e., magic squares with six square entries in a particular configuration. We then impose a second condition to create a system of equations whose solution, if it exists, would yield a non-Bremner magic square of seven distinct square entries; a lack of solution, however, would prove the nonexistence of a magic square of squares.

1. Introduction

The magic squares of squares problem is an open problem in recreational mathematics that asks whether a 3×3 magic square can contain nine distinct square integer entries. This problem, formally posed in 1984, is intriguing not just for its simplicity of explanation, but also for the fact that magic squares of squares exist for dimensions $n \geq 4$ [Boy05]. While there have been many attempts at the magic squares of squares problem [Boya], both computational and theoretical, it seems rather unlikely that a magic square of squares exists, unless its entries are astronomically large. As such, we posit the following conjecture, for which we will, in this paper, provide the necessary background and tools to allow the reader to investigate the magic squares of squares problem, should they choose to do so.

Conjecture 1.1. A primitive magic square of squares does not exist.

See Definition 2.4 for the notion of primitivity. Instead of focusing directly on magic squares of squares, we take an approach similar to [Bre01] where we focus on "simpler" creatures, namely, radical magic diamonds and, eventually, magic diamonds of squares; see Definitions 2.11 and 2.10, respectively. The reason for doing so is, as of writing, there has only been one magic square containing sevens distinct square entries, that being Bremner's square [Bre99]. Bremner's square is not a magic diamond of squares, so if a magic diamond of squares exists, then it will be a non-Bremner square with seven square entries. On the other hand, if a magic diamond of squares does not exist, then a magic square of squares cannot exist since every magic square of squares is a magic diamond of squares. These two cases correspond, respectively, to open problems 2 and 1 of [Boyb]. Curiously, while the existence or nonexistence of a magic diamond of squares has yet to be shown over $\mathbb{Z}$, one was discovered over $\mathbb{Z}/2^{59}\mathbb{Z}$ in [PTZ15].

Organization. We assume that magic squares of squares exist. After introducing the relevant definitions in Section 2, we then develop a series of magic square invariants and representations in Section 3. This will allow us to introduce Lucas form, which every magic squares of squares must be able to be written as.

Theorem 1.2. *Any magic square of squares can be written in Lucas form:*

$$\begin{pmatrix} \gamma+\alpha & \gamma-\alpha-\beta & \gamma+\beta \\ \gamma-\alpha+\beta & \gamma & \gamma+\alpha-\beta \\ \gamma-\beta & \gamma+\alpha+\beta & \gamma-\alpha \end{pmatrix}$$

where

$$\alpha = 2x_1x_2, \qquad \beta = 2x_3x_4, \qquad x_1^2+x_2^2 = \gamma = x_3^2+x_4^2.$$

However, starting with a magic square in Lucas form, some further conditions must be satisfied for said square to be a magic square of squares, those being:

Theorem 1.3. *The necessary conditions for an magic squares of squares in Lucas form are*

$$x_1^2+x_2^2 = X^2 = x_3^2+x_4^2, \qquad (x_1+x_2)^2 = y_1^2+y_2^2,$$
$$(x_1-x_2)^2 = y_3^2+y_4^2, \qquad y_1y_2 = x_3x_4 = y_3y_4.$$

This research began as the author's WELP:U project with Wolfram and continued as an independent project at Caltech.

Through some further manipulation of Lucas form and its necessary conditions, we can show that every magic square of squares can be written in Euclid form.

Theorem 1.4. *Any magic square of squares can be written in Euclid form:*

$$\begin{pmatrix} a & b-\Delta & W+\Delta \\ b+\Delta & W & a-\Delta \\ W-\Delta & a+\Delta & b \end{pmatrix}$$

where

$$a = k_1^2(m^2+n^2)^2, \qquad b = k_2^2(p^2+q^2)^2, \qquad W = \frac{a+b}{2},$$

$$4k_1^2mn(m^2-n^2) = \Delta = 4k_2^2pq(p^2-q^2).$$

Euclid form is stronger than Lucas form in the sense that, without the following necessary conditions, it guarantees any magic square which can be written in Euclid form contains at least six square entries, those being on the diamond component. Here, we also shift to focusing on primitive magic squares of squares to begin directly working towards Conjecture 1.1.

Theorem 1.5. *The necessary conditions for a primitive magic squares of squares in Euclid form are W is square along with*

$$m > n > 0, \qquad \gcd(m,n) = 1, \qquad m \not\equiv n \bmod 2,$$

$$p > q > 0, \qquad \gcd(p,q) = 1, \qquad p \not\equiv q \bmod 2,$$

$$k_1, k_2 > 0, \qquad \gcd(k_1,k_2) = 1, \qquad k_1, k_2 \not\equiv 0 \bmod 2.$$

Euclid form also allows us to view the magic squares of squares problem more geometrically. After showing some further results on what values the entries of a magic square of squares can take, we get the following:

Theorem 1.6. *Every primitive magic square of squares can be written in Lucas form:*

$$\begin{pmatrix} W+\Delta^\star & W-\Delta^\star-\Delta & W+\Delta \\ W-\Delta^\star+\Delta & W & W+\Delta^\star-\Delta \\ W-\Delta & W+\Delta^\star+\Delta & W-\Delta^\star \end{pmatrix}$$

where $W \equiv 1 \bmod 24$ is square and $\Delta, \Delta^\star \equiv 0 \bmod 24$ are congrua.

We can also consider the behavior for the ending digits of the entries of a magic square of squares.

Theorem 1.7. *The only possible primitive magic square of squares representations over $\mathbb{Z}/10\mathbb{Z}$, up to rotation/reflection, are*

$$\begin{pmatrix} 1 & 9 & 5 \\ 9 & 5 & 1 \\ 5 & 1 & 9 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{pmatrix}.$$

Similar to the work of [Rob96], our work will culminate in three equivalent restatements of Conjecture 1.1, those being the following:

Conjecture 1.8. The center entry W of Euclid form cannot be square.

Conjecture 1.9. The hypotenuses of two distinct Pythagorean triangles with equal area cannot be in a square arithmetic progression with middle value W an integer.

Conjecture 1.10. Assuming a primitive magic squares of squares exists, it cannot contain an entry divisible by 5, and the ending digits of every entry cannot be the same.

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2. Preliminary Definitions

Definition 2.1. A 3×3 matrix, or **square**, has the **magic property** if all its rows, columns, and diagonals sum to the same value K, called the **magic constant**; a square with this property is called a **magic square**. Specifically, given the representation in Figure 1, a magic square satisfies the row sums

$$a+b+c=K, \qquad d+e+f=K, \qquad g+h+i=K,$$

the column sums

$$a+d+g=K, \qquad b+e+h=K, \qquad c+f+i=K,$$

and the diagonal sums

$$a+e+i=K, \qquad c+e+g=K.$$

With this notation, we call a, c, g, i the **corner entries**, b, d, f, h the **middle entries**, and e the **center entry**.

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

Figure 1. A representation of a square.

Remark 2.2. Throughout the paper, we assume the entries of any given square are integers unless specified otherwise.

Definition 2.3. A magic square is **nontrivial** if all its entries are distinct. Furthermore, a magic square is **primitive** if all its entries are coprime, i.e., the GCD of all entries is 1.

Remark 2.4. By definition, a primitive magic square is nontrivial. Also, as with primitive Pythagorean triples, a primitive magic square generates a family of magic squares via scalar multiplication.

Definition 2.5. Given a square, the **X-component** and **O-component** of the square refer, respectively, to the representations in Figure 2.

$$\begin{pmatrix} a & - & c \\ - & e & - \\ g & - & i \end{pmatrix} \qquad \begin{pmatrix} - & b & - \\ d & e & f \\ - & h & - \end{pmatrix}$$

Figure 2. The X-component (left) and O-component (right) of a square.

Definition 2.6. Given a square, the **hourglass component** and **diamond component** of the square refer, respectively, to the representations in Figure 3.

$$\begin{pmatrix} a & b & c \\ - & e & - \\ g & h & i \end{pmatrix} \qquad \begin{pmatrix} a & b & - \\ d & e & f \\ - & h & i \end{pmatrix}$$

Figure 3. The hourglass component (left) and diamond component (right) of a square.

Remark 2.7. In general, an hourglass component can be thought of as an X-component with a pair of opposing middle entries. Similarly, a diamond component can be thought of as an O-component with a pair of opposing corner entries. Note that we take the word "opposing" to always mean across the center entry.

Definition 2.8. A **magic hourglass** is a square whose hourglass component has the magic property. Similarly, a **magic diamond** is a square whose diamond component has the magic property.

Remark 2.9. A magic square is both a magic hourglass and a magic diamond, but the converse does not necessarily hold.

Definition 2.10. A square has the **of squares property** if all its entries are perfect squares; a magic square with this property is called a **magic square of squares** (MSS). We can extend this definition to similarly define a **magic hourglass of squares** (MHS) and a **magic diamond of squares** (MDS).

Definition 2.11. A square has the **radical property** if all its entries except the center entry are perfect squares; a magic square with this property is called a **radical magic square** (RMS). We can extend this definition to similarly define a **radical magic hourglass** (RMH) and a **radical magic diamond** (RMD).

Remark 2.12. An RMS is both an RMH and an RMD, but the converse does not necessarily hold. Also, for any given square, the radical property is equivalent to the of squares property over $\mathbb{Z}$ adjoin the square root of its center entry.

3. Magic Square Invariants

Lemma 3.1. *The magic property is invariant under matrix rotation and reflection. Furthermore, the magic property is invariant under rational scalar addition and multiplication while the magic constant, however, is not.*

Proof. Clearly, any matrix rotation and reflection permutes the entries of a magic square while preserving the row, column, and diagonal sums. Also, scalar addition and multiplication apply globally, i.e., to each entry of a given magic square. The mechanical verification is left to the reader. □

Remark 3.2. In Lemma 3.1, we specified scalar addition and multiplication over $\mathbb{Q}$ due to direct relevance to the paper. However, invariance under these transformations also holds over $\mathbb{R}$ and even $\mathbb{Z}/n\mathbb{Z}$ for any n. In fact, a magic square with $\mathbb{Z}$-entries is still a magic square over $\mathbb{Z}/n\mathbb{Z}$.

Lemma 3.3. *The **first invariant** of magic squares is*

$$K = 3e.$$

In words, each row, column, and diagonal sum to three times the center entry.

Proof. Since

$$a+e+i=K, \qquad b+e+h=K, \qquad c+e+g=K,$$

then combining equations implies that

$$a+b+c+3e+i+h+g=3K.$$

Furthermore, since

$$a+b+c=K, \qquad i+h+g=K,$$

then

$$K = 3e.$$

□

Lemma 3.4. *The **second invariant** of magic squares is*

$$a+i=2e, \qquad b+h=2e, \qquad c+g=2e, \qquad d+f=2e.$$

In words, both opposing corner entries and opposing middle entries sum to two times the center entry.

Proof. Since

$$a+e+i=K, \qquad b+e+h=K, \qquad c+e+g=K, \qquad d+e+f=K,$$

then the first invariant implies that

$$a+i=2e, \qquad b+h=2e, \qquad c+g=2e, \qquad d+f=2e.$$

□

Lemma 3.5. *The **third invariant** of magic squares is*

$$b=g+i-e, \qquad d=c+i-e, \qquad f=a+g-e \qquad h=a+c-e.$$

In words, the middle entries are the sum of the two opposing corner entries minus the center entry.

Proof. Since

$$b+e+h=g+h+i, \qquad d+e+f=c+f+i,$$
$$d+e+f=a+d+g, \qquad b+e+h=a+b+c,$$

then

$$b=g+i-e, \qquad d=c+i-e, \qquad f=a+g-e \qquad h=a+c-e.$$

□

Lemma 3.6. *The **fourth invariant** of magic squares is*

$$2a=f+h, \qquad 2c=d+h, \qquad 2g=b+f, \qquad 2i=b+d.$$

In words, the corner entries are half of the sum of two opposing middle entries.

Proof. By the first and second invariant,

$$b+h+c+g+d+f+2a=2K+2a, \qquad a+i+b+h+d+f+2c=2K+2c,$$
$$a+i+b+h+d+f+2g=2K+2g, \qquad b+h+c+g+d+f+2i=2K+2i,$$

so since

$$a+b+c=K, \qquad c+f+i=K, \qquad g+h+i=K, \qquad a+d+g=K,$$

then

$$2a=f+h, \qquad 2c=d+h, \qquad 2g=b+f, \qquad 2i=b+d.$$

□

Lemma 3.7. *A magic square written in terms of its X-component is said to be in **X-form**. Similarly, a magic square written in terms of its O-component is said to be in **O-form**. Any magic square can be written in X-form and O-form, respectively, as*

$$\begin{pmatrix} a & g+i-e & c \\ c+i-e & e & a+g-e \\ g & a+c-e & i \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \frac{f+h}{2} & b & \frac{d+h}{2} \\ d & e & f \\ \frac{b+f}{2} & h & \frac{b+d}{2} \end{pmatrix}.$$

Proof. We can write a magic square in X-form (resp. O-form) by applying the third invariant (resp. fourth invariant) to its X-component (resp. O-component). □

Remark 3.8. By Lemma 3.7, we only need four datum to generate a magic square, namely any solutions to

$$x_1+x_2=x_3+x_4.$$

We can take this a step further by generating any magic square with a single rational number.

Proposition 3.9. *A magic square written with center entry zero is said to be in **center form**. Any magic square with top left entry a, top right entry c, and center entry e can be written in center form as*

$$\begin{pmatrix} \iota & -(1+\iota) & 1 \\ 1-\iota & 0 & -(1-\iota) \\ -1 & 1+\iota & -\iota \end{pmatrix}$$

where

$$\iota=\frac{a-e}{c-e}.$$

Proof. Consider a magic square in X-form. Since magic squares are invariant under scalar addition, we can subtract through by e to arrive at representation

$$\begin{pmatrix} a-e & (g-e)+(i-e) & c-e \\ (c-e)+(i-e) & 0 & (a-e)+(g-e) \\ g-e & (a-e)+(c-e) & i-e \end{pmatrix}.$$

By the second invariant,

$$(g-e)=-(c-e), \qquad (i-e)=-(a-e),$$

so the previous representation becomes

$$\begin{pmatrix} a-e & -(c-e)-(a-e) & c-e \\ (c-e)-(a-e) & 0 & (a-e)-(c-e) \\ -(c-e) & (a-e)+(c-e) & -(a-e) \end{pmatrix}.$$

Since magic squares are invariant under scalar multiplication, we can divide through by $c-e$. □

4. Magic Squares of Squares

Lemma 4.1. *The solutions of the Diophantine equations*

$$X^2+Y^2=2Z^2 \qquad \textit{and} \qquad x^2+y^2=z^2$$

are in a bijection via

$$(X,Y,Z)\mapsto\left(\frac{X+Y}{2},\frac{X-Y}{2},Z\right) \qquad \textit{and} \qquad (x,y,z)\mapsto(x+y,x-y,z).$$

Proof. For first equation, we have that $2 \mid X^2+Y^2$, so X,Y are of the same parity. Hence, $2 \mid X+Y$ and $2 \mid X-Y$, so the expressions $\frac{X+Y}{2}$ and $\frac{X-Y}{2}$ are integers. Thus, the first map is well-defined into $\mathbb{Z}$. Bijectivity follows via composition:

$$(X,Y,Z)\mapsto\left(\frac{X+Y}{2},\frac{X-Y}{2},Z\right)\mapsto\left(\frac{X+Y}{2}+\frac{X-Y}{2},\frac{X+Y}{2}-\frac{X-Y}{2},Z\right)=(X,Y,Z)$$

and

$$(x,y,z)\mapsto(x+y,x-y,z)\mapsto\left(\frac{(x+y)+(x-y)}{2},\frac{(x+y)-(x-y)}{2},Z\right)=(x,y,z).$$

□

Theorem 4.2. *Any MSS can be written in* ***Lucas form****:*

$$\begin{pmatrix} \gamma+\alpha & \gamma-\alpha-\beta & \gamma+\beta \\ \gamma-\alpha+\beta & \gamma & \gamma+\alpha-\beta \\ \gamma-\beta & \gamma+\alpha+\beta & \gamma-\alpha \end{pmatrix}$$

where

$$\alpha=2x_1x_2, \qquad \beta=2x_3x_4, \qquad x_1^2+x_2^2=\gamma=x_3^2+x_4^2.$$

Proof. By Lemma 3.7, we can write any given MSS in X-form:

$$\begin{pmatrix} X_1^2 & X_2^2+X_4^2-X^2 & X_3^2 \\ X_2^2+X_3^2-X^2 & X^2 & X_1^2+X_4^2-X^2 \\ X_4^2 & X_1^2+X_3^2-X^2 & X_2^2 \end{pmatrix}$$

where, by the second invariant,

$$X_1^2+X_2^2=2X^2=X_3^2+X_4^2.$$

By Lemma 4.1, we can write the noncenter entries as

$$X_1=x_1+x_2, \qquad X_2=x_1-x_2, \qquad X_3=x_3+x_4, \qquad X_4=x_3-x_4$$

where the x_1,x_2,x_3,x_4 satisfy

$$x_1^2+x_2^2=X^2=x_3^2+x_4^2.$$

Thus, the previous representation becomes

$$\begin{pmatrix} (x_1+x_2)^2 & (x_1-x_2)^2+(x_3-x_4)^2-X^2 & (x_3+x_4)^2 \\ (x_1-x_2)^2+(x_3+x_4)^2-X^2 & X^2 & (x_1+x_2)^2+(x_3-x_4)^2-X^2 \\ (x_3-x_4)^2 & (x_1+x_2)^2+(x_3+x_4)^2-X^2 & (x_1-x_2)^2 \end{pmatrix}.$$

Since

$$(x_1\pm x_2)^2\mp 2x_1x_2=X^2=(x_3\pm x_4)^2\mp 2x_3x_4,$$

then we have

$$\begin{pmatrix} X^2+2x_1x_2 & X^2-2x_1x_2-2x_3x_4 & X^2+2x_3x_4 \\ X^2-2x_1x_2+2x_3x_4 & X^2 & X^2+2x_1x_2-2x_3x_4 \\ X^2-2x_3x_4 & X^2+2x_1x_2+2x_3x_4 & X^2-2x_1x_2 \end{pmatrix}.$$

□

Corollary 4.3. *Any MSS can be written in center form:*

$$\begin{pmatrix} \iota & -(1+\iota) & 1 \\ 1-\iota & 0 & -(1-\iota) \\ -1 & 1+\iota & -\iota \end{pmatrix}$$

where

$$\iota = \frac{x_1 x_2}{x_3 x_4}.$$

Proof. This follows immediately from Proposition 3.9 and Theorem 4.2. □

Remark 4.4. Rewriting Lucas form as

$$\begin{pmatrix} (x_1+x_2)^2 & (x_1-x_2)^2 - 2x_3x_4 & (x_3+x_4)^2 \\ (x_1-x_2)^2 + 2x_3x_4 & x_1^2+x_2^2 & (x_1+x_2)^2 - 2x_3x_4 \\ (x_3-x_4)^2 & (x_1+x_2)^2 + 2x_3x_4 & (x_1-x_2)^2 \end{pmatrix},$$

we see that the condition

$$x_1^2 + x_2^2 = X^2 = x_3^2 + x_4^2$$

alone is not enough to, in general, guarantee that the middle entries are square. However, via the bijectivity of Lemma 4.1, this condition is sufficient to generate every magic square whose X-component consists of square entries. Note that Lucas form is named after the square discovered by Édouard Lucas in [Luc91].

Theorem 4.5. *The necessary conditions for an MSS in Lucas form are*

$$x_1^2 + x_2^2 = X^2 = x_3^2 + x_4^2, \qquad (x_1+x_2)^2 = y_1^2 + y_2^2,$$
$$(x_1-x_2)^2 = y_3^2 + y_4^2, \qquad y_1y_2 = x_3x_4 = y_3y_4.$$

Proof. Given the Lucas form representation in Remark 4.4, we automatically have

$$x_1^2 + x_2^2 = X^2 = x_3^2 + x_4^2.$$

For a square in Lucas form to be an MSS, we require that

$$(x_1+x_2)^2 + 2x_3x_4 = Y_1^2, \qquad (x_1+x_2)^2 - 2x_3x_4 = Y_2^2,$$
$$(x_1-x_2)^2 + 2x_3x_4 = Y_3^2, \qquad (x_1-x_2)^2 - 2x_3x_4 = Y_4^2,$$

for some integers Y_1, Y_2, Y_3, Y_4. Combining equations, we have

$$(x_1+x_2)^2 = \frac{Y_1^2+Y_2^2}{2}, \qquad (x_1-x_2)^2 = \frac{Y_3^2+Y_4^2}{2}, \qquad Y_1^2 - Y_2^2 = 4x_3x_4 = Y_3^2 - Y_4^2.$$

Applying Lemma 4.1 to the first two equations implies that

$$(x_1+x_2)^2 = y_1^2 + y_2^2, \qquad (x_1-x_2)^2 = y_3^2 + y_4^2,$$

where the y_1, y_2, y_3, y_4 satisfy

$$Y_1 = y_1 + y_2, \qquad Y_2 = y_1 - y_2, \qquad Y_3 = y_3 + y_4, \qquad Y_4 = y_3 - y_4.$$

Thus,

$$(y_1+y_2)^2 - (y_1-y_2)^2 = 4x_3x_4 = (y_3+y_4)^2 - (y_3-y_4)^2,$$

which simplifies to

$$y_1y_2 = x_3x_4 = y_3y_4.$$

□

Corollary 4.6. *The necessary conditions for an RMS in Lucas form are*

$$x_1^2 + x_2^2 = x_3^2 + x_4^2, \qquad (x_1+x_2)^2 = y_1^2 + y_2^2,$$
$$(x_1-x_2)^2 = y_3^2 + y_4^2, \qquad y_1y_2 = x_3x_4 = y_3y_4.$$

Proof. We simply relax the condition in Proposition 4.5 that the center entry is a square. □

Corollary 4.7. *The necessary conditions for an MDS in Lucas form are*

$$x_1^2 + x_2^2 = X^2, \qquad (x_1+x_2)^2 = y_1^2 + y_2^2,$$
$$(x_1-x_2)^2 = y_3^2 + y_4^2, \qquad y_1y_2 = y_3y_4.$$

Proof. We simply relax the condition in Corollary 4.6 that the corner entries involving x_3, x_4 are square. □

Corollary 4.8. *The necessary conditions for an RMD in Lucas form are*

$$(x_1+x_2)^2 = y_1^2+y_2^2, \qquad (x_1-x_2)^2 = y_3^2+y_4^2, \qquad y_1y_2 = y_3y_4.$$

Proof. We simply relax the condition in Corollary 4.7 that the center entry is a square. □

5. Geometric Implications

Remark 5.1. We can rewrite the necessary conditions for an RMD in Lucas form as

$$a_1^2+b_1^2 = c_1^2, \qquad a_2^2+b_2^2 = c_2^2, \qquad a_1b_1 = a_2b_2,$$

where

$$c_1 = x_1+x_2, \qquad c_2 = x_1-x_2,$$

and the y_1, y_2, y_3, y_4 are simply relabeled. With this labeling, it is now easy to geometrically interpret a solution to this system of Diophantine equations: Every RMD corresponds to a pair of Pythagorean triangles with equal area.

Example 5.2. The Pythagorean triples $(12, 35, 37)$ and $(20, 21, 29)$ correspond to two Pythagorean triangles with area 210. Taking

$$c_1 = 37, \qquad c_2 = 29, \qquad y_1y_2 = 420,$$

as in Remark 5.1, we then have

$$x_1 = 33, \qquad x_2 = 4, \qquad x_3x_4 = 420,$$

as in Theorem 4.5. Thus, substituting $\alpha = 264, \beta = 840, \gamma = 1105$ into Lucas form, we get the RMD

$$\begin{pmatrix} \gamma+\alpha & \gamma-\alpha-\beta & \gamma+\beta \\ \gamma-\alpha+\beta & \gamma & \gamma+\alpha-\beta \\ \gamma-\beta & \gamma+\alpha+\beta & \gamma-\alpha \end{pmatrix} = \begin{pmatrix} 37^2 & 1^2 & 1945 \\ 41^2 & 1105 & 23^2 \\ 265 & 47^2 & 29^2 \end{pmatrix}.$$

This RMD corresponds to center form with $\iota = \frac{11}{35}$:

$$\begin{pmatrix} \iota & -(1+\iota) & 1 \\ 1-\iota & 0 & -(1-\iota) \\ -1 & 1+\iota & -\iota \end{pmatrix} = \begin{pmatrix} \frac{11}{35} & -\frac{46}{35} & 1 \\ \frac{24}{35} & 0 & -\frac{24}{35} \\ -1 & \frac{46}{35} & -\frac{11}{35} \end{pmatrix}.$$

Theorem 5.3. *Any MSS can be written in* ***Euclid form****:*

$$\begin{pmatrix} a & b-\Delta & W+\Delta \\ b+\Delta & W & a-\Delta \\ W-\Delta & a+\Delta & b \end{pmatrix}$$

where

$$a = k_1^2(m^2+n^2)^2, \qquad b = k_2^2(p^2+q^2)^2, \qquad W = \frac{a+b}{2},$$
$$4k_1^2mn(m^2-n^2) = \Delta = 4k_2^2pq(p^2-q^2).$$

Proof. Via Euclid's formula (Theorem 12.1 of [Bur07]), we can parametrize the conditions

$$(x_1+x_2)^2 = y_1^2+y_2^2, \qquad (x_1-x_2)^2 = y_3^2+y_3^2$$

for an MSS in Lucas form by

$$y_1 = k_1(m^2-n^2), \qquad y_2 = 2k_1mn, \qquad x_1+x_2 = k_1(m^2+n^2),$$
$$y_3 = k_2(p^2-q^2), \qquad y_4 = 2k_2pq, \qquad x_1-x_2 = k_2(p^2+q^2).$$

Thus, the condition

$$y_1y_2 = x_3x_4 = y_3y_4$$

becomes

$$2k_1^2mn(m^2-n^2) = x_3x_4 = 2k_2^2pq(p^2-q^2).$$

Also, by Lemma 4.1, the condition

$$x_1^2+x_2^2 = X^2 = x_3^2+x_4^2$$

is equivalent to

$$\frac{(x_1+x_2)^2+(x_1-x_2)^2}{2} = X^2 = \frac{(x_3+x_4)^2+(x_3-x_4)^2}{2}.$$

Rewriting Lucas form as

$$\begin{pmatrix} (x_1+x_2)^2 & (x_1-x_2)^2-2x_3x_4 & X^2+2x_3x_4 \\ (x_1-x_2)^2+2x_3x_4 & X^2 & (x_1+x_2)^2-2x_3x_4 \\ X^2-2x_3x_4 & (x_1+x_2)^2+2x_3x_4 & (x_1-x_2)^2 \end{pmatrix},$$

we have via substitution

$$\begin{pmatrix} k_1^2(m^2+n^2)^2 & k_2^2(p^2+q^2)^2-2x_3x_4 & X^2+2x_3x_4 \\ k_2^2(p^2+q^2)^2+2x_3x_4 & X^2 & k_1^2(m^2+n^2)^2-2x_3x_4 \\ X^2-2x_3x_4 & k_1^2(m^2+n^2)^2+2x_3x_4 & k_2^2(p^2+q^2)^2 \end{pmatrix}.$$

□

Remark 5.4. Via the parametrization in Theorem 5.3, which is bijective in the sense that every Pythagorean triple can be generated (although perhaps with multiple representations), the necessary conditions for an RMD in Lucas form become the single equation

$$k_1^2 mn(m^2-n^2) = k_2^2 pq(p^2-q^2),$$

which is inherent to the definition of Euclid form. This is apparent when we rewrite Euclid form as

$$\begin{pmatrix} k_1^2(m^2+n^2)^2 & k_2^2(p^2-2pq-q^2)^2 & A_1 \\ k_2^2(p^2+2pq-q^2)^2 & A_2 & k_1^2(m^2-2mn-n^2)^2 \\ A_3 & k_1^2(m^2+2mn-n^2)^2 & k_2^2(p^2+q^2)^2 \end{pmatrix}$$

where

$$A_1 = \frac{k_1^2(m^2+2mn-n^2)^2+k_2^2(p^2+2pq-q^2)^2}{2}, \qquad A_2 = \frac{k_1^2(m^2+n^2)^2+k_2^2(p^2+q^2)^2}{2},$$

$$A_3 = \frac{k_1^2(m^2-2mn-n^2)^2+k_2^2(p^2-2pq-q^2)^2}{2}.$$

Note that the value Δ is geometrically significant because it is a *congruum* [OEI15], which is never square by Fermat's right triangle theorem (Theorem 12.5 of [Bur07]). As such, we now see that every RMD corresponds to a pair of triples in square arithmetic progression with the same congruum, namely,

$$(a-\Delta, a, a+\Delta), \qquad (b-\Delta, b, b+\Delta).$$

Overall, every MSS corresponds to a triple of triples in square arithmetic progression with the same congruum, namely,

$$(a-\Delta, a, a+\Delta), \qquad (b-\Delta, b, b+\Delta), \qquad (W-\Delta, W, W+\Delta).$$

Equivalently, we can write

$$(a-\Delta, a, a+\Delta), \qquad (b-\Delta, b, b+\Delta),$$

$$a-W=\Delta^\star, \qquad W-b=\Delta^\star,$$

for some congruum $\Delta^\star$. Since the values $\sqrt{a}, \sqrt{b}$ of Euclid form are, by construction, the values c_1, c_2 in Remark 5.2, then we can interpret the arithmetic progression for an MSS geometrically as in Figure 4.

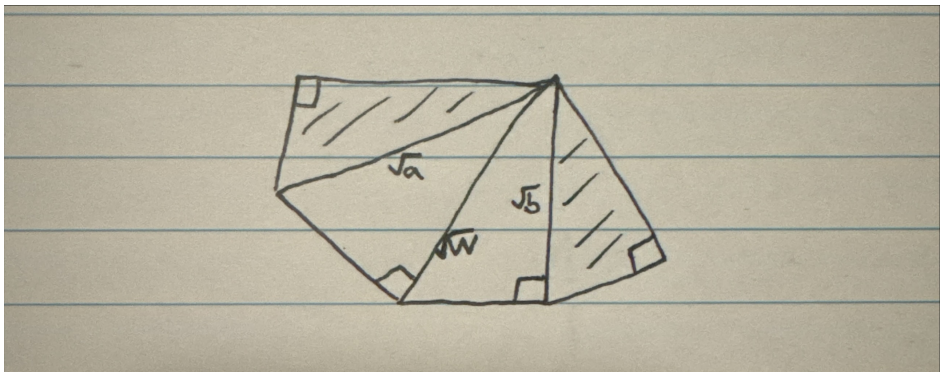


FIGURE 4. A geometric problem.

Example 5.5. The first nontrivial solution to

$$k_1^2 mn(m^2 - n^2) = k_2^2 pq(p^2 - q^2)$$

is

$$k_1 = 1, \qquad m = 6, \qquad n = 1, \qquad k_2 = 1, \qquad p = 5, \qquad q = 2.$$

Taking

$$a = 1369, \qquad b = 841, \qquad W = 1105, \qquad \Delta = 840,$$

as in Euclid form, we get the same RMD as in Example 5.2:

$$\begin{pmatrix} a & b-\Delta & W+\Delta \\ b+\Delta & W & a-\Delta \\ W-\Delta & a+\Delta & b \end{pmatrix} = \begin{pmatrix} 37^2 & 1^2 & 1945 \\ 41^2 & 1105 & 23^2 \\ 265 & 47^2 & 29^2 \end{pmatrix}.$$

We now see that the triples $(23^2, 37^2, 47^2)$ and $(1^2, 29^2, 41^2)$ are in arithmetic progression with congruum 840. Also, the triples $(41^2, 1945, 47^2)$ and $(1^2, 265, 23^2)$ are in arithmetic progression with congruum 264. The next three RMDs which can be generated via this process are

$$\begin{pmatrix} 197^2 & 31^2 & 36265 \\ 151^2 & 25345 & 167^2 \\ 14425 & 223^2 & 109^2 \end{pmatrix}, \qquad \begin{pmatrix} 401^2 & 73^2 & 130945 \\ 263^2 & 99025 & 359^2 \\ 67105 & 439^2 & 193^2 \end{pmatrix}, \qquad \begin{pmatrix} 365^2 & 213^2 & 118781 \\ 291^2 & 99125 & 337^2 \\ 79469 & 391^2 & 255^2 \end{pmatrix}.$$

Remark 5.6. Via the parametrization in Theorem 5.3, the necessary conditions for an MDS in Lucas form become the system of equations

$$k_1^2 mn(m^2 - n^2) = k_2^2 pq(p^2 - q^2), \qquad k_1^2(m^2 + n^2)^2 + k_2^2(p^2 + q^2)^2 = 2X^2.$$

Hence, the existence/nonexistence of an MDS depends solely one whether, given a solution to the first equation, said solution can ever satisfy the second equation for an arbitrary integer X. Note that Bremner's square in Figure 5 is not an MDS.

$$\begin{pmatrix} 373^2 & 289^2 & 565^2 \\ 360721 & 425^2 & 23^2 \\ 205^2 & 527^2 & 222121 \end{pmatrix}$$

FIGURE 5. Bremner's square.

6. Modular Sudoku

Remark 6.1. So far, we have intentionally neglected to impose any restrictions on the system of equations defining Euclid form. Doing so has allowed us to develop a general theory of MSSs. However, to address the magic squares of squares problem more directly, we want to turn our attention to primitive MSSs.

Proposition 6.2. *Every entry of a primitive MSS is* 1 mod 2.

Proof. We can write any given MSS in Euclid form:

$$\begin{pmatrix} a & b-\Delta & W+\Delta \\ b+\Delta & W & a-\Delta \\ W-\Delta & a+\Delta & b \end{pmatrix}$$

where

$$a = k_1^2(m^2+n^2)^2, \qquad b = k_2^2(p^2+q^2)^2, \qquad W = \frac{a+b}{2},$$

$$4k_1^2mn(m^2-n^2) = \Delta = 4k_2^2pq(p^2-q^2).$$

By definition of W, we must have $2 \mid a+b$ for W to be an integer, so a, b are of the same parity. For the sake of contradiction, assume that both are even. Since a, b are squares, then $2^2 \mid a$ and $2^2 \mid b$. Similarly, by definition of Δ, we must have $2^2 \mid \Delta$ for Δ to be an integer. Thus, we can write

$$a = 2^2a', \qquad b = 2^2b', \qquad \Delta = 2^2\Delta',$$

for some integers a', b', Δ'. Thus, Euclid form becomes

$$\begin{pmatrix} 2^2a' & 2^2(b'-\Delta) & W+2^2\Delta' \\ 2^2(b'+\Delta') & W & 2^2(a'-\Delta') \\ W-2^2\Delta' & 2^2(a'+\Delta') & 2^2b' \end{pmatrix}$$

where

$$W = \frac{a+b}{2} = 2^2\left(\frac{a'+b'}{2}\right).$$

We must have $2 \nmid a'+b'$, else $2^2 \mid W$ and, subsequently, the entire MSS is divisible by 2^2, contradicting primitivity. Thus, $a'+b'$ is odd, so $\frac{a'+b'}{2}$ is a reduced rational fraction, which can never be square over $\mathbb{Q}$. It follows that $2^2\left(\frac{a'+b'}{2}\right)$ can never square over $\mathbb{Z}$, so W is not square. This contradicts the assumption that our magic square has the of squares property, so a, b must be odd. Since a, b are squares, then we can write

$$a = (2j+1)^2, \qquad b = (2k+1)^2,$$

for some integers j, k. Thus,

$$W = 2(j^2+j+k^2+k)+1,$$

so W is odd. Since Δ is even, then all entries of our MSS are odd. □

Corollary 6.3. *Every entry of a primitive MSS is* 1 mod 8.

Proof. By Proposition 6.2, the entries of an MSS are odd squares. Over $\mathbb{Z}/8\mathbb{Z}$, these entries can take values

$$1^2 \mapsto 1, \qquad 3^2 \mapsto 1, \qquad 5^2 \mapsto 1, \qquad 7^2 \mapsto 1.$$

Hence, all entries are 1 mod 8. □

Remark 6.4. Proposition 6.2 is an alternative proof of Theorem 1.1 in [Rab03].

Theorem 6.5. *The necessary conditions for a primitive MSS in Euclid form are W is square along with*

$$m > n > 0, \qquad \gcd(m,n) = 1, \qquad m \not\equiv n \bmod 2,$$

$$p > q > 0, \qquad \gcd(p,q) = 1, \qquad p \not\equiv q \bmod 2,$$

$$k_1, k_2 > 0, \qquad \gcd(k_1,k_2) = 1, \qquad k_1, k_2 \not\equiv 0 \bmod 2.$$

Proof. By definition of Euclid's formula, we automatically have

$$m > n > 0, \qquad \gcd(m, n) = 1, \qquad m \not\equiv n \bmod 2,$$
$$p > q > 0, \qquad \gcd(p, q) = 1, \qquad p \not\equiv q \bmod 2.$$

Since Euclid's formula assumes positive Pythagorean triples, then k_1, k_2 are positive. Also, if k_1, k_2 are not coprime, then our MSS will be divisible by their common divisors, a contradiction to primitivity. Similarly, either k_1, k_2 are even, then our MSS will have even entries, a contradiction to Proposition 6.2. Thus, we have

$$k_1, k_2 > 0, \qquad \gcd(k_1, k_2) = 1, \qquad k_1, k_2 \not\equiv 0 \bmod 2.$$

□

Lemma 6.6. *Every entry of a primitive MSS is* 1 mod 3.

Proof. From Euclid form, we have the squares a, b, W satisfying

$$W = \frac{a+b}{2},$$

which, over $\mathbb{Z}/3\mathbb{Z}$, can take values

$$0^2 \mapsto 0, \qquad 1^2 \mapsto 1, \qquad 2^2 \mapsto 1.$$

If $W \equiv 0 \bmod 3$, then $a + b \equiv 0 \bmod 3$, which is only possible if $a, b \equiv 0 \bmod 3$. The first invariant implies that $a \pm \Delta, b \pm \Delta, W \pm \Delta \equiv 0 \bmod 3$, so every entry of our MSS is divisible by 3, contradicting primitivity. Thus, $W \equiv 1 \bmod 3$, and the same argument implies that all entries are 1 mod 3. □

Remark 6.7. Lemma 6.6 is an alternative proof of Theorem 10 in [Mor15].

Proposition 6.8. *Every entry of a primitive MSS is* 1 mod 24.

Proof. Let η be an entry in our MSS. By Corollary 6.3 and Lemma 6.5, we have

$$\eta \equiv 1 \bmod 3, \qquad \eta \equiv 1 \bmod 8.$$

Since $\gcd(3, 8) = 1$, then we can use the Chinese remainder theorem (Theorem 4.8 of [Bur07]): From

$$8\eta_1 \equiv 1 \bmod 3, \qquad 3\eta_2 \equiv 1 \bmod 8,$$

we have

$$\eta_1 = 2, \qquad \eta_2 = 3.$$

Thus,

$$\eta = 8\eta_1 + 3\eta_2 = 25 \equiv 1 \bmod 24.$$

□

Remark 6.9. Proposition 6.8 is an alternative proof of Lemma 1 in [PTZ15].

Theorem 6.10. *Every primitive MSS can be written in Lucas form:*

$$\begin{pmatrix} W + \Delta^\star & W - \Delta^\star - \Delta & W + \Delta \\ W - \Delta^\star + \Delta & W & W + \Delta^\star - \Delta \\ W - \Delta & W + \Delta^\star + \Delta & W - \Delta^\star \end{pmatrix}$$

where $W \equiv 1 \bmod 24$ *is square and* $\Delta, \Delta^\star \equiv 0 \bmod 24$ *are congrua.*

Proof. The Lucas form representation follows immediately by substituting

$$a = W + \Delta^\star, \qquad b = W - \Delta^\star,$$

as in Remark 5.4, into Euclid from. By Proposition 6.6, we have that all entries are 1 mod 24, so Δ must be 0 mod 24. □

Lemma 6.11. *The only possible primitive MSS representations over* $\mathbb{Z}/5\mathbb{Z}$*, up to rotation/reflection, are*

$$\begin{pmatrix} 0 & 1 & 4 \\ 4 & 0 & 1 \\ 1 & 4 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix}.$$

Proof. From Euclid form, we have the squares a, b, W satisfying

$$W = \frac{a+b}{2},$$

which, over $\mathbb{Z}/5\mathbb{Z}$, can take values

$$0^2 \mapsto 0, \qquad 1^2 \mapsto 1, \qquad 2^2 \mapsto 4, \qquad 3^2 \mapsto 4, \qquad 4^2 \mapsto 1.$$

If $W \equiv 0 \bmod 5$, then $a + b \equiv 0 \bmod 5$, which is possible if either $a, b \equiv 0 \bmod 5$ or one of a, b is $1 \bmod 5$ while the other is $4 \bmod 5$. Thus, we have the partial representations

$$\begin{pmatrix} 0 & - & - \\ - & 0 & - \\ - & - & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & - & - \\ - & 0 & - \\ - & - & 4 \end{pmatrix}, \qquad \begin{pmatrix} 4 & - & - \\ - & 0 & - \\ - & - & 1 \end{pmatrix}.$$

The only valid completions of these representations, which adhere to both primitivity and the first invariant, are

$$\begin{pmatrix} 0 & 1 & 4 \\ 4 & 0 & 1 \\ 1 & 4 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 4 & 1 \\ 1 & 0 & 4 \\ 4 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 1 & 0 \\ 1 & 0 & 4 \\ 0 & 4 & 1 \end{pmatrix}.$$

If $W \equiv 1 \bmod 5$, then $a + b \equiv 2 \bmod 5$, which is only possible if $a, b \equiv 1 \bmod 5$. Similarly, if $W \equiv 4 \bmod 5$, then $a + b \equiv 3 \bmod 5$, which is only possible if $a, b \equiv 4 \bmod 5$. These cases correspond, respectively, to the partial representations

$$\begin{pmatrix} 1 & - & - \\ - & 1 & - \\ - & - & 1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & - & - \\ - & 4 & - \\ - & - & 4 \end{pmatrix}.$$

The only valid completions of these representations, which adhere to the first invariant, are

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix}.$$

□

Theorem 6.12. *The only possible primitive MSS representations over $\mathbb{Z}/10\mathbb{Z}$, up to rotation/reflection, are*

$$\begin{pmatrix} 5 & 1 & 9 \\ 9 & 5 & 1 \\ 1 & 9 & 5 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{pmatrix}.$$

Proof. Let η be an entry in our MSS. By Proposition 6.2 and Lemma 6.8, we have three possibilities:

$$\eta \equiv 1 \bmod 2, \qquad \eta \equiv 0 \bmod 5;$$

$$\eta \equiv 1 \bmod 2, \qquad \eta \equiv 1 \bmod 5;$$

$$\eta \equiv 1 \bmod 2, \qquad \eta \equiv 4 \bmod 5.$$

Since $\gcd(2, 5) = 1$, then we can use the Chinese remainder theorem to give a map from $\mathbb{Z}/5\mathbb{Z}$ to $\mathbb{Z}/10\mathbb{Z}$ by

$$0 \mapsto 5, \qquad 1 \mapsto 1, \qquad 4 \mapsto 9.$$

Thus, the desired representations follow via substitution on

$$\begin{pmatrix} 0 & 1 & 4 \\ 4 & 0 & 1 \\ 1 & 4 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix}.$$

□

Remark 6.13. Since working over $\mathbb{Z}/10\mathbb{Z}$ is the same as working with last digits in $\mathbb{Z}$, then Theorem 6.10 implies that an MSS either has exactly one diagonal divisible by 5 or all of its entries end in the same digit.

7. Conclusion

We now consider some reformulations of Conjecture 1.1, which could prove to be easier in approach and even resolution. If an MDS does not exist, then an MSS likewise does not exist, so Remark 5.6 implies the following:

Conjecture 7.1. The center entry W of Euclid form cannot be square.

Also, if the geometry in Figure 4 is not allowed, then we have the following:

Conjecture 7.2. The hypotenuses of two distinct Pythagorean triangles with equal area cannot be in a square arithmetic progression with middle value W an integer.

Lastly, if we can eliminate the three possible cases in Theorem 6.12, then we have the following:

Conjecture 7.3. Assuming a primitive MSS exists, it cannot contain an entry divisible by 5, and the ending digits of every entry cannot be the same.

Conjecture 7.3 seems interesting because perhaps one can construct maps from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{Z}/m\mathbb{Z}$ as in Theorem 6.12 to eliminate one or both of its conditions.

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